
Do Vision-Language Models Recognize Product Attributes Beyond Catalog Style?

A Controlled Study of Attribute Recognition and Style Robustness in Fashion E-commerce

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Abstract

We propose a controlled study of vision-language models for fashion e-commerce. Our central question is whether strong model performance on product understanding tasks reflects genuine recognition of semantic attributes or reliance on presentation cues such as catalog style. Using the DeepFashion dataset, we evaluate two vision-language models, CLIP and BLIP, on attribute recognition for three interpretable attributes: category, color, and sleeve length. We then compare performance across catalog-like and less curated images to test robustness under style shift. In addition, we include a small prompt perturbation analysis to examine attribute sensitivity under controlled mismatches. Our goal is to provide a feasible and interpretable evaluation of whether current vision-language models recognize product attributes beyond standardized e-commerce presentation.

1 Research Question and Motivation

Our project studies the following question:

When a vision-language model predicts fashion product attributes correctly, is it recognizing the visual attribute itself, or is it relying on presentation cues such as catalog style?

This question is important because many e-commerce images follow highly standardized visual patterns, including clean backgrounds, consistent lighting, and similar poses. Under such conditions, a model may appear to understand product attributes such as category, color, or sleeve length while actually exploiting superficial regularities in image presentation. If so, strong benchmark performance may overstate true visual understanding.

This issue matters both scientifically and practically. From a research perspective, it helps clarify whether multimodal models learn semantic visual concepts or benefit from dataset artifacts. From an application perspective, real e-commerce systems must handle not only polished catalog photos but also less curated consumer-style images. A model that performs well only under standardized presentation may not be reliable in realistic deployment.

2 Dataset

We use **DeepFashion** as the primary dataset for this project. DeepFashion provides clothing images with category and attribute annotations and includes images with varying levels of visual curation, making it suitable for testing both attribute recognition and style robustness.

To keep the project feasible, we focus on a controlled subset with reliable labels for three attributes:

- **category** (e.g., dress, shirt, jacket)
- **color** (e.g., black, white, red, blue)
- **sleeve length** (e.g., sleeveless, short sleeve, long sleeve)

Our preprocessing will include:

- filtering invalid or corrupted images
- selecting balanced samples for each attribute class
- normalizing labels into a consistent vocabulary
- constructing a style-based split between catalog-like and less curated images

This controlled setup allows us to evaluate whether attribute recognition remains stable when presentation style changes.

3 Models

We compare two pretrained vision-language models:

- **CLIP** as a baseline model
- **BLIP** as a stronger comparison model

Both models map images and text into compatible representation spaces and can be used for image-text matching without additional training. We use publicly available pretrained implementations through standard libraries such as Hugging Face Transformers.

No full fine-tuning will be performed. All experiments are based on zero-shot or prompt-based inference, which keeps the project computationally lightweight and feasible within a limited time frame.

4 Methodology

This project is a controlled evaluation study rather than a new model design project. We focus on two main experiments and one small auxiliary analysis.

4.1 Experiment 1: Controlled Attribute Recognition

The first experiment evaluates whether models can recognize semantic product attributes through image-text matching.

For each image, we construct a small set of candidate prompts corresponding to possible values of a given attribute. For example:

- color: “a red dress”, “a blue dress”, “a black dress”
- category: “a dress”, “a shirt”, “a jacket”
- sleeve length: “a sleeveless top”, “a short-sleeve shirt”, “a long-sleeve shirt”

The model computes similarity scores between the image and each candidate prompt. The predicted label is the prompt with the highest score.

This experiment provides a controlled baseline for determining which attributes are recognized reliably and which remain difficult.

4.2 Experiment 2: Style Robustness Under Domain Shift

The second experiment tests whether attribute recognition remains stable when image presentation style changes.

We divide the dataset into two subsets:

- **catalog-like images**
- **less curated / consumer-like images**

We then evaluate the same attribute recognition task on both subsets and compare the results. The main quantity of interest is the performance drop from catalog-like to less curated images.

If the official dataset split does not directly provide this distinction, we will define a proxy split based on metadata or visual characteristics and document the criterion clearly.

This experiment is central to our study because a substantial performance drop would suggest that the model depends in part on presentation shortcuts rather than purely on semantic attribute understanding.

4.3 Auxiliary Analysis: Prompt Perturbation Sanity Check

As a small controlled auxiliary analysis, we will test whether models are sensitive to incorrect attribute descriptions.

For a limited set of images, we will compare the similarity score of a correct prompt with the score of slightly perturbed incorrect prompts. For example:

- correct: “a red dress”
- wrong color: “a blue dress”
- wrong category: “a red jacket”

Rather than treating this as a full third experiment, we use it as a sanity check on attribute sensitivity. This provides additional evidence about whether the models meaningfully distinguish correct from incorrect descriptions.

5 Evaluation Metrics

We evaluate the project using the following metrics:

- **overall accuracy**
- **per-attribute accuracy**
- **confusion matrices**
- **domain gap**: performance difference between catalog-like and less curated subsets
- **prompt margin**: similarity difference between correct and perturbed prompts in the auxiliary analysis

We will also include qualitative error analysis to inspect common failure modes and determine which attributes are most affected by style variation.

6 Implementation Plan

The project will be implemented using:

- Python
- PyTorch
- Hugging Face Transformers

- NumPy / Pandas
- Matplotlib

Because the study uses pretrained models without full training, it can be completed with modest computational resources. The implementation pipeline consists of:

1. preprocessing DeepFashion and selecting a controlled subset
2. constructing attribute-based text prompts
3. running CLIP and BLIP inference
4. computing image-text similarity scores
5. evaluating accuracy and robustness metrics
6. visualizing results and inspecting errors

7 Expected Outcomes

We expect that both models will perform reasonably well on common attributes such as category and color under catalog-like conditions. However, we hypothesize that:

- performance will vary across attribute types
- sleeve length may be less robust than category
- performance will drop on less curated images
- the stronger model may improve absolute accuracy but still remain sensitive to presentation style

These findings would suggest that strong performance on standardized e-commerce images does not necessarily imply full semantic understanding of product attributes.

8 Contribution

This project provides a feasible and controlled evaluation framework for studying whether vision-language models recognize product attributes beyond standardized catalog presentation. By restricting the study to one dataset, two models, three attributes, and two main experiments, we aim to produce interpretable results within a manageable scope.

Although lightweight, the project addresses a meaningful research question about shortcut reliance in multimodal systems and can be extended to additional datasets and models in future work.

References

- [1] Ziwei Liu, Ping Luo, Shi Qiu, Xiaogang Wang, and Xiaoou Tang. DeepFashion: Powering robust clothes recognition and retrieval with rich annotations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1096–1104, 2016.
- [2] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021.
- [3] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pages 12888–12900, 2022.